4. $M = P.(1+i)^n$

$5000 = P.(1+0,0182)^5$

$P = \frac{5000}{(1+0,0182)^5}$

$P = \frac{5000}{(1,0182)^5}$

$P = \frac{5000}{1,09464}$

$P = 4564,66$

5. $M = P.(1+i.t)$

$13P = P.(1+i.t)$

$1,3 = 1 + i.t$

$i.t = 1,3 - 1$

$i.t = 0,3$

$i.20 = 0,3$

$i = \frac{0,3}{20}$

$i = 0,015$

$i_a = 0,015.12$

$i_a = 0,18$

6. $M = P + P.i.t$

$3P = P + P.0,05.t$

$3 = 1 + 0,05.t$

$0,05.t = 2$

$t = \frac{2}{0,05}$

$t = 40$

7. $J = P.i.t$

$J = 192.0,30.8$

$P = \frac{J}{i.t}$

$P = \frac{192}{0,30.\frac{8}{12}}$

$P = \frac{192}{0,20}$

$P = 960$

8. $2.000 + 2200 + 2420 + 2662$

$2,928,20 = 14,210,20 - 10000$

$4210,20$

$J = P.i.t$

$4210,20 = 10000.i.\frac{30}{30}$

$i = \frac{4210,20}{10000}$

$i = 0,4210$

$i = 4,21\ \%$

9. $M_1 = P_1 + P_1.i.t$

$M_1 = 5000 + 5000.0,02.t$

$M_2 = 4000 + 4000.0,0375.t$

$5000 + 100t = 4000 + 150t$

$100t - 150t = 4000 - 5000$

$-50t = -1000$

$t = 20$

10. $M_C = 2000 + 2000.0,02.m$

$M_P = 1850 + 1850.0,03.m$

$M_C = M_P = 2000 + 40m = 1850 + 55,5m$

$-15,5m = -150$

$m = \frac{150}{15,5}$

$m = 9,6774$

$9 + 6 + 7 + 7 + 4 = 33$

11. $A(1+0,05t)\quad B(1+0,04t)\quad C = (1+0,03t)$

$A + B + C = 48902$

$A(1+0,05.1) = 18720(1+0,03.1)$

$A(1,05) = 18720(1,03)$

$A = \frac{18720.1,03}{1,05}$

$A = 18,358,86$

12. $1,5V + 0,36V = 600$

$1,86V = 600$

$V = \frac{600}{1,86}$

$V = 322,58$

$P + 0.6.322,58 = 1000$

$1000 - 0,6.322,58$

$P = 798,71$

$798,71 + 322,58 = 1121,29$